OXIDATION BEHAVIOUR OF NIOBIUM–CHROMIUM ALLOYS

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SUMMARY

An examination of the properties of chromium niobate indicates the possibility of producing an oxidation-resistant niobium–chromium alloy. A series of such alloys has been prepared and subjected to oxidation tests. Tests on one of these alloys in different atmospheres reveal a considerably higher oxidation rate in air than in oxygen. A possible mechanism is suggested to account for this phenomenon.

INTRODUCTION

A necessary condition for self protection of a niobium-rich alloy against oxidation at elevated temperatures is the formation of an adherent self-healing oxide coating on its surface. The structure of this oxide should provide as complete a barrier as possible to the diffusion of oxygen inwards and metal cations outwards. The adhesion of the oxide is determined firstly by the volume-ratio of the oxide and underlying alloy and secondly by their relative thermal expansions. For good adhesion, the maximum permissible volume ratio resulting from application of the Pilling–Bedworth relationship is 1.15. However, the value for chromic oxide on chromium is 2.07 and silica on silicon 1.88, and since both oxides are protective, this proposal seems rather pessimistic and a volume ratio of 2 may well be acceptable if other factors are favourable. The high temperature form of niobium pentoxide, on the other hand, has a volume ratio of 2.7, which explains why the oxide skin flakes away from niobium as soon as appreciable quantities of this highest oxide have formed.

An important factor concerning adhesion of the oxide, especially when subjected to thermal shocks is its thermal expansion coefficient. This should be as near to that of the parent metal as possible.

Finally, the best diffusion barrier is likely to be provided by the more complex double or multiple oxide structures, as is found for example with the spinel-type structures formed on nickel–chromium alloys and austenitic heat-resisting steels.

INVESTIGATION OF DOUBLE OXIDES

As a preliminary study various powder mixtures of Nb_2O_5 and other metal oxides were prepared, pressed and sintered, and the resultant material examined by X-ray diffraction. From this study three double oxides were found which it was thought

might provide good diffusion barriers. These three double oxides are:

$CrNbO_4$
$NiNb_2O_6$
ZrO_2–Nb_2O_5 (solid solution)

Bars of these combinations, suitable for use in a conventional dilatometer, were prepared. The appropriate proportions of powder were thoroughly mixed by tumbling for 20 h, pressed at room temperature in a steel die with cavity dimensions $\frac{1}{4}$ in. × $\frac{1}{4}$ in. × $1\frac{1}{2}$ in. and the resulting bars sintered for 15 h at 1,300°C in air. Dilatometer curves over the temperature range 20°–1,000°C showed the following mean thermal expansion coefficients.

$CrNbO_4$	5.9 p.p.m./°C
$NiNb_2O_6$	3.0 p.p.m./°C
ZrO_2–Nb_2O_5	4.5 p.p.m./°C

The curve for $CrNbO_4$ was not, however, smooth, but showed a sudden steepening just below 800°C. To ascertain whether this was due to a phase change, the material was studied in a high temperature X-ray diffraction camera. No new phase appeared to be formed. The mean thermal expansion coefficient of $CrNbO_4$ measured by this method over the temperature range 20°–840°C was 5.1 p.p.m./°C, which is in reasonable agreement with the dilatometer figure averaged over 20°–1,000°C.

It appeared, therefore, that on the basis of its thermal expansion, $CrNbO_4$ was the most promising of the three double oxides studied, although the expansion coefficient of the parent alloy had the considerably higher value of 9 p.p.m./°C.

The reported structure of the niobate $CrNbO_4$ has been confirmed at Caswell Laboratories. Its tetragonal atomic arrangement contains one molecule per unit cell of dimensions $a = 4.64$ Å, $c = 3.013$ Å at room temperature (*i.e.* 65 Å^3 volume). The volume ratio with respect to its theoretical parent alloy (50–50 atomic % Cr–Nb), was calculated to be 2.1. To estimate the extent to which it might hinder diffusion, a comparison was made with a spinel structure.

	Vol. per molecule (Å^3)	*Number of atoms/molecule*	
		Metal	*Oxygen*
A typical spinel structure ($MeO.Mt_2O_3$)	70	3	4
$CrNbO_4$	65	2	4

From this the niobate structure appears to be reasonably promising as a diffusion barrier.

On all three criteria, thermal expansion, volume ratio and structure, there should then be some prospect of a useful degree of self-protection in a 50–50 atomic % Cr–Nb alloy, the theoretical parent of $CrNbO_4$.

PREPARATION AND OXIDATION OF NIOBIUM–CHROMIUM ALLOYS

Although the 50–50 atomic % Nb–Cr binary was of immediate interest, initial oxidation results from this material suggested that lower chromium contents down

to say 20 atomic %, should be explored to see how far reasonable oxidation properties could be achieved without a severe brittleness problem. Accordingly, samples containing 50, 45, 40, 35, 30 and 25 atomic % of chromium were prepared. The preparation of the alloys was carried out in an atmosphere of argon, using the cold hearth R.F. melting technique developed at CASWELL[1]. This melting technique is very similar to arc-melting except that R.F. heating is used instead of an arc. The specimens were melted a second time after inversion of the specimen, in order to ensure complete melting. This was followed by heating to 1,500°C under a vacuum of $5 \cdot 10^{-5}$ mm Hg for 30 min in order to facilitate some degree of homogenisation.

Chemical analysis checks on the final composition of some of these alloys showed that, unlike sintering processes involving chromium, negligible loss of this metal occurred when this method of preparation was used.

Oxidation tests on these niobium–chromium binaries were carried out using a Stanton Recording Balance in conjunction with a platinum furnace. The design was such that the atmosphere could be varied as required. The following test procedure was employed. The alloy was heated in the platinum furnace until a temperature of 1,100°C was reached, during which time a stream of argon flowed over the specimen at the rate of 0.4 l/min, thus preventing any oxidation. On reaching a temperature of 1,100°C a stream of dried oxygen flowing at 0.4 l/min, was substituted for the argon. Weight changes were recorded continuously by the balance. The normal oxidation tests were of 17 h duration. A longer confirmatory test of 65 h was carried out where necessary.

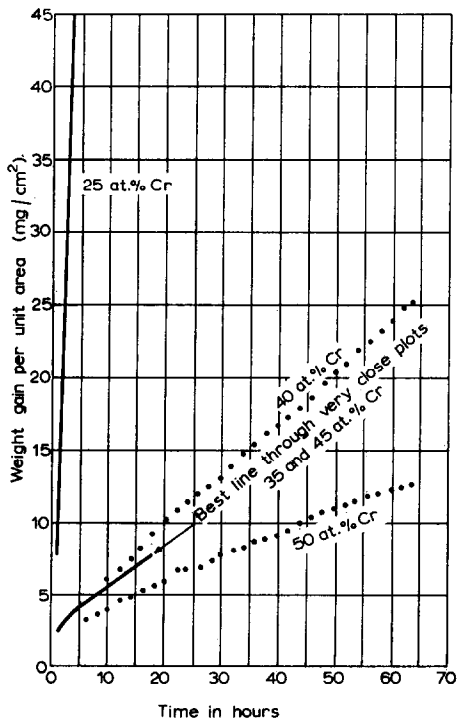


Fig. 1. Oxidation of various niobium–chromium alloys in oxygen at 1,100°C.

The curves of weight gain per unit area against time under these conditions, are presented in Fig. 1. In order to ascertain the type of oxidation law operating, the results are plotted logarithmically in Fig. 2. This plot will be linear with a slope of $1/x$ if a single power law of the following type is obeyed:

$$W^x = at + c$$

where W = weight gain/unit area
t = time
a and c are constants

The results obtained for alloys with chromium contents between 35 and 50 atomic %, give values of x between 1 and 2. The type of oxidation law is thus intermediate between linear and parabolic. The rapid oxidation rate of the 75–25 atomic % Nb–Cr binary suggests that for a simple binary alloy, the minimum chromium content for reasonable oxidation resistance is about 35 atomic %.

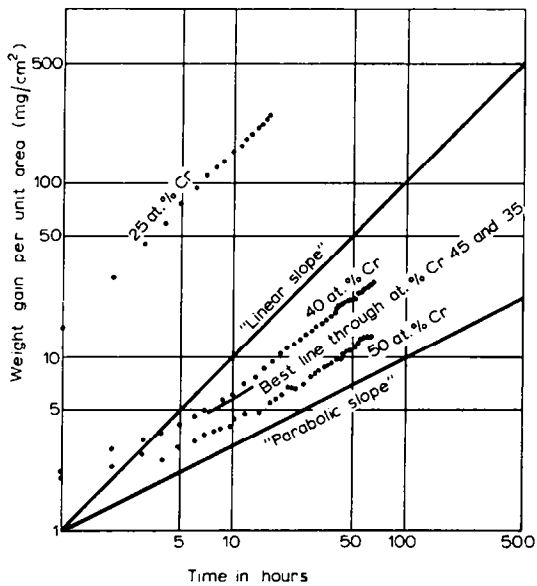


Fig. 2. Oxidation of various niobium–chromium alloys in oxygen at 1,100°C.

X-ray diffraction analysis of the oxide film from the alloy containing 50 atomic % chromium, which is the theoretical parent of $CrNbO_4$, showed that this double oxide was formed during oxidation although appreciable amounts of H-type (high temperature form) Nb_2O_5 were also present. The presence of this excess Nb_2O_5 is undoubtedly due to preferential oxidation of niobium which would be expected from a consideration of the free energy of formation of Nb_2O_5 and Cr_2O_3 (—373 kcal/mole for Nb_2O_5 and —180 kcal/mole for Cr_2O_3 at 1,000°C)[2].

For further study a composition of 65–35 atomic % Nb–Cr was chosen. Any increase in the chromium content does not improve oxidation resistance appreciably, but is likely to accentuate the problem of brittleness associated with chromium-containing alloys.

PRELIMINARY OXIDATION OF A 65–35 ATOMIC % Nb–Cr ALLOY

Although more severe oxidation was expected to occur in pure oxygen than in air, this assumption has been shown to be incorrect in the case of this alloy. Fig. 3 shows weight gain/cm² plotted against time for oxidation tests carried out in air and oxygen. Both tests were conducted at 1,100°C in the thermobalance with gas flows of 0.4 l/min. After 17 h the weight gain/cm² in air was four to five times that in oxygen and the ratio after 60 h was even greater. To attempt to explain this phenomenon the effect of nitrogen on the oxidation of the 65–35 atomic % Nb–Cr binary was investigated in more detail.

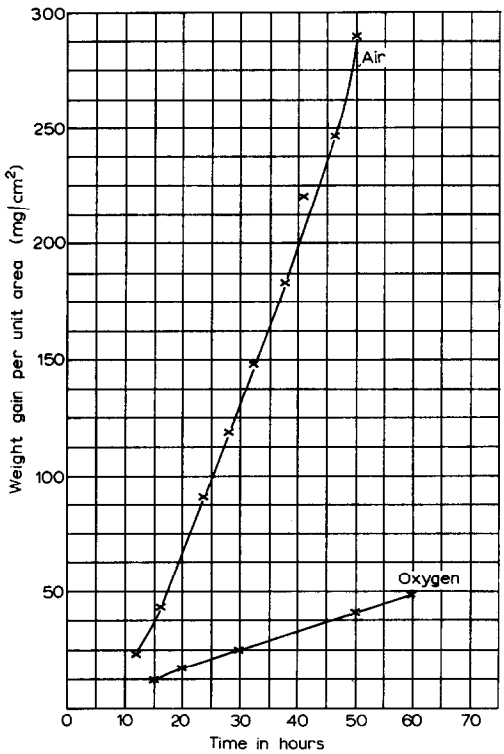


Fig. 3. Oxidation of a 65–35 at. % niobium–chromium alloy in air and oxygen at 1,100°C.

THE EFFECT OF NITROGEN ON THE OXIDATION RATE OF A 65–35 ATOMIC % Nb–Cr ALLOY

Experimental

A series of oxidation tests was carried out in oxygen–nitrogen mixtures containing between 10 and 96 vol. % of nitrogen. For all tests hearth-melted specimens were used. The tests were carried out in the thermobalance for 17 h at a temperature of 1,100°C using hearth-melted specimens. The composition of the gas mixture was controlled by needle-valves and the total rate of flow maintained at 0.4 l/min. The gases were mixed and dried with anhydrone before entering the reaction chamber. The reaction of pure nitrogen with the 65–35 atomic % Nb–Cr alloy for 17 h at

1,100°C was performed in a specially constructed furnace to eliminate the possibility of oxygen contamination due to back diffusion of air. High purity nitrogen dried by anhydrone and passed over heated niobium to remove traces of oxygen was used.

Two tests were carried out in the thermobalance involving pretreatment of hearth-melted specimens at 1,100°C in nitrogen, directly followed by treatment in pure oxygen at the same temperature. One test consisted of a 17-h run in nitrogen followed by a further 40 h in oxygen, whilst in the second one, the run in nitrogen was limited to 2 h, again followed by 40 h in oxygen. The rate of flow of the gases was 0.4 l/min in each case.

A further oxidation test using the same temperature and gas flow conditions was performed in the thermobalance using a 50–50 vol. % mixture of argon and oxygen. The test was carried out to confirm that the behaviour described was associated with the presence of nitrogen.

One of the factors causing the increased oxidation rate of 65–35 atomic % Nb–Cr alloy when nitrogen is introduced into an oxygen atmosphere may well be the ability of nitrogen to diffuse into the alloy more readily than oxygen. This possibility has been investigated by micro-hardness measurements.

Four hearth-melted specimens were used for these experiments, two being treated in oxygen and two in nitrogen. In each case the treatment was carried out for a period of 17 h at 1,100°C using a gas flow of 0.4 l/minute. The specimens were then sectioned and the surface carefully polished. A line of micro-hardness measurements was taken perpendicular to the surface of the alloy. The measurements were made using a Leitz Miniload Hardness Tester with a 300 g load.

Results

The results of the oxidation tests carried out in various oxygen–nitrogen mixtures are presented in Fig. 4. The graph shows that the weight gain rises sharply at either end of the plot to a value of about 40 mg/cm² when the proportion of either nitrogen or oxygen is very small in the gas mixture. This weight gain then scarcely varies for any other proportion of the mixed gases. From these results it would appear that the

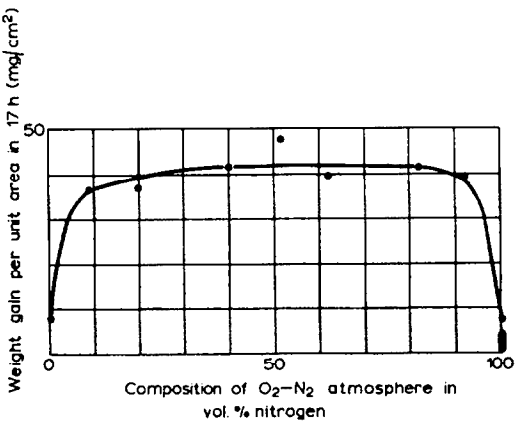


Fig. 4. Reaction of a 65–35 at. % niobium–chromium alloy with different oxygen–nitrogen mixtures at 1,100°C.

nitrogen is acting as a catalyst, only a very small amount being needed to increase the rate of reaction by four or five times.

The test in pure nitrogen shows a weight gain comparable to that in pure oxygen (the value is plotted in Fig. 4). The surface of the specimen appeared to be barely tarnished, but a micrograph (Fig. 5) of a section through the nitrided alloy shows that a definite film was formed. The existence of this film was confirmed by microhardness measurements which showed the presumed nitride layer to be considerably harder than the alloy itself. It was possible to scrape a small amount of "nitrided" film from the surface of the specimen and this was examined by X-ray diffraction. However, it was not possible to interpret the X-ray patterns, neither niobium nor chromium nitride being present. It is probable that a complex mixed niobium–chromium nitride was formed.

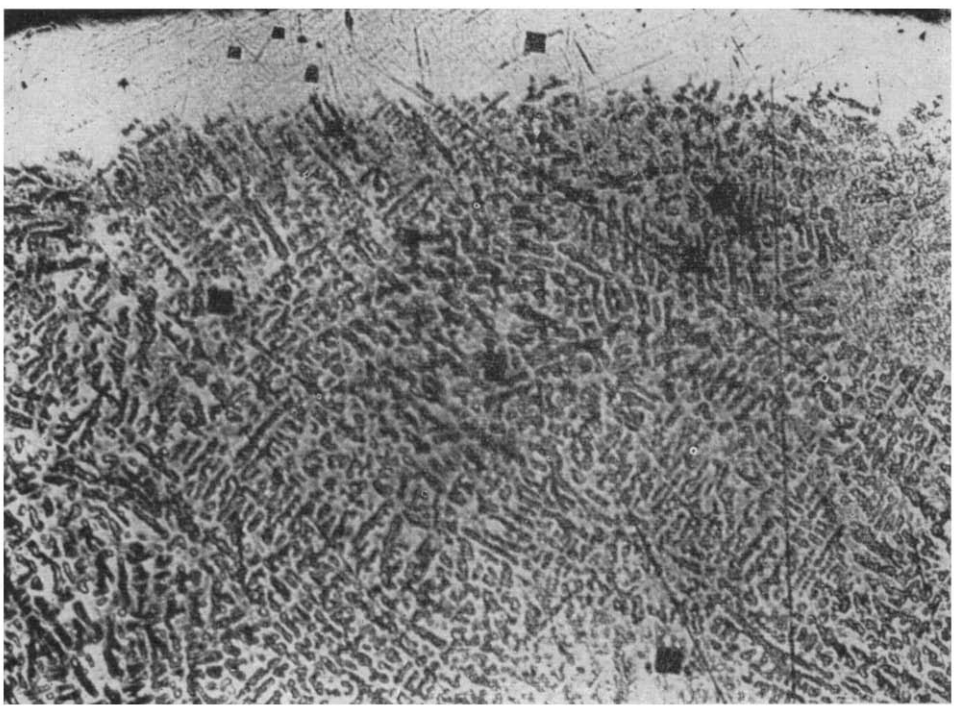


Fig. 5. 65–35 at. % Nb–Cr binary after treating with nitrogen at 1,100°C for 17 h. Etched aqua regia + 20% HF, showing a thin compact nitride layer. The microhardness indentations show this layer to be harder than the alloy itself. (× 200)

The results of the oxidation tests carried out after pretreatment in nitrogen, are shown in Fig. 6. The oxidation curves given commence at the time the oxygen was introduced. In the first test (17 h pretreatment in nitrogen), curve A shows that the initial reaction rate in oxygen was 72 mg/cm^2 in the first 17 h (*i.e.* much above the rate normally observed in pure oxygen). However, after 21 h there was a distinct and marked decrease in the reaction rate, as can be seen from the graph. No such marked change is apparent in curve B (2 h pretreatment in nitrogen), although a slight decrease in reaction rate is perceptible after about 15 h.

The oxidation rate in a 50–50 vol. % mixture of oxygen–argon proved to be of the same order as that in pure oxygen, thus confirming that a reduction in the partial pressure of oxygen does not increase the oxidation rate.

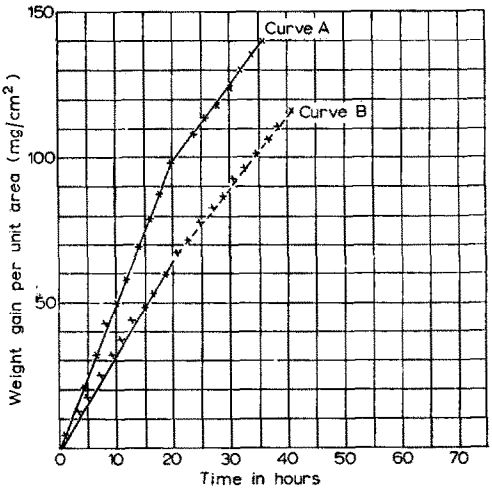


Fig. 6. Reaction of a 65–35 at. % niobium–chromium alloy at 1,100°C following pretreatment with nitrogen.

Fig. 7 shows the result of hardness measurements taken on the specimens oxidised in oxygen and air. Although the curves are somewhat variable, they do show that the nitrogen penetration exceeds that of oxygen.

A general hardness figure for a 65–35 atomic % Nb–Cr alloy in the "as-melted" condition is 900–1,000 V.P.N. For the alloys treated in oxygen the hardness figure falls to this value at depths of 50 μ (curve A) and 100 μ (curve B), whereas for the

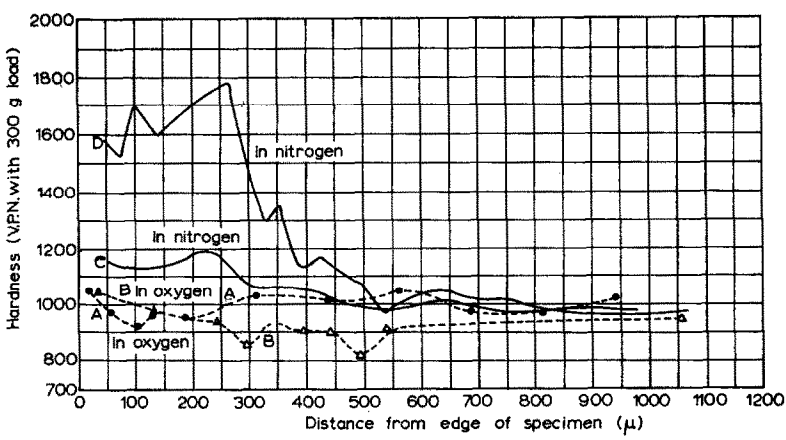


Fig. 7. Hardness of a 65–35 at. % niobium–chromium alloy after treatment in oxygen and nitrogen.

alloys treated in nitrogen the hardness exceeds 1,000 V.P.N. until depths of 450 μ (curve C) and 550 μ (curve D) are reached.

Discussion

The above results show quite conclusively that the increase of oxidation rate in oxygen–nitrogen mixtures over that in pure oxygen is due to the catalytic action of nitrogen. It has been shown by hardness measurements that nitrogen is capable of faster diffusion into this alloy than is oxygen. This would suggest that the "nitride" is formed before the oxide. It has been shown by X-ray diffraction of the film formed by air oxidation that no nitride is present. The film consists only of $CrNbO_4$ and H-type Nb_2O_5. Thus it would appear that the conversion of nitride to oxide is very fast.

The results of oxidation following pretreatment in nitrogen (Fig. 6) may be explained as follows. Pretreatment in nitrogen for 17 h allows nitrogen to diffuse into the alloy to a considerable depth and this facilitates subsequent attack by oxygen, thus giving a fast reaction rate (curve A, Fig. 6). However, when oxidation has proceeded so far that the complete shell of "nitrided" alloy has been oxidised, no further catalysis takes place, and a substantial decrease in the rate of reaction results. This state of affairs is apparently reached in 20 h in this instance. The rate of reaction in the latter stages (after 20 h) is still greater than in pure oxygen, but this is probably due to traces of "nitride" still remaining, perhaps as a result of regeneration of nitrogen on oxidation of the "nitride".

It was in an attempt to confirm this hypothesis that the second test was undertaken, *i.e.* 2 h pretreatment in nitrogen. It was thought that by considerably reducing the pretreatment time in nitrogen, the changes in reaction rate would occur after a shorter time. Although the overall reaction rate in oxygen (curve B) was much lower than in the previous test, no marked changes in the rate occurred, but a very slight decrease after some 15 h was observed. It may be noted, however, that the complete slope of curve B is the same as that of the latter portion of curve A, which seems to suggest that the diffusion of nitrogen into the alloy during the 2 h pretreatment was only slight. This being the case, no very marked change in rate of reaction with oxygen would be expected.

The exact nature of the catalytic action of the nitrogen is not known, but two suggestions can be put forward. The first is that the fast prediffusion of the nitrogen allows much easier and faster diffusion of oxygen, thus giving a faster rate of reaction. The following hypothesis is preferred, however. The nitrogen diffuses through the alloy, forming a complex "nitride" which is immediately oxidised, this conversion of "nitride" to oxide being much faster than the simple oxidation of alloy to oxide. Two possible oxidation mechanisms can, therefore, occur depending on the atmosphere.
Oxidation in pure oxygen

$$(Nb, Cr) + O_2 \xrightarrow{\text{slow}} CrNbO_4 + Nb_2O_5$$

Oxidation in nitrogen and oxygen

$$(Nb, Cr) + N_2 \longrightarrow (Nb, Cr - N) + O_2 \xrightarrow{\text{fast}} CrNbO_4 + Nb_2O_5 + N_2$$

It is in fact possible that the complete reaction of alloy via "nitride" to oxide is faster than alloy to oxide direct.

It can be seen from Fig. 3 that the reaction rate of the 65–35 atomic % Nb–Cr alloy with air at 1,100°C increases with time. A possible explanation is that the thin semi-protective film primarily formed on the alloy surface becomes more and more cracked and porous as the oxidation progresses.

CONCLUSIONS

The results show beyond doubt that the presence of nitrogen has a very adverse effect on the oxidation rate of a 65–35 atomic % Nb–Cr (the effect probably occurring with a whole series of Nb–Cr alloys). As neither niobium nor chromium has a faster oxidation rate in air than in oxygen, it must be concluded that it is the presence in the Nb–Cr alloy of either the solid solution (chromium in niobium) or the compound ($NbCr_2$) or both, which enables nitrogen catalysis of the oxidation process to occur.

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